

Prateek Sinha

Chennai, India | ps826105@gmail.com | github.com/CzPhantom10 | +91 914211703

EDUCATION

SRM Institute of Science and Technology

Kattankulathur, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

- CGPA: 8.32
- Active participant in Hackathons and LLM based Application Projects

EXPERIENCE

AI Engineer

Jan 2024 – Present

SRM Institute of Science and Technology

- Fine-tuned **GPT-2** to create **LawGPT**, a domain-specific legal LLM for Indian legal Q&A and judgment summarization.
- Built a **transformer-based language model from scratch** using PyTorch for foundational NLP experimentation.
- Created a **YOLOv8**-based fire & smoke detection system for real-time safety surveillance using OpenCV and Streamlit.
- Implemented a **Retrieval-Augmented Generation (RAG)** model integrating vector search for context-aware AI responses.

PROJECTS

APEX AI SURVIVAL ARENA | *PyTorch, Pygame, NumPy*

Sept 2025

- Implemented a multi-agent Deep Q-Network (DQN) survival simulation with 8+ trained agent personalities in a procedurally generated 12×12 grid featuring seasons, resource & temperature mechanics, and complex survival actions.
- Designed a persistent betting economy with real-time leaderboard and statistics, polished UI/UX (particle effects & animations), and cross-platform distribution via Docker and standalone builds.

LawGPT | *Python, PyTorch, GPT-2, CUDA*

May 2025

- Fine-tuned GPT-2 on Indian legal datasets (IPC, CrPC, judgments) for legal question answering
- Implemented LoRA and custom tokenizers for efficient training

QuestMint | *Python, Python Flask, Groq API, Solidity, React, CSS3, Ethers.js*

Apr 2025

- Developed an AI + blockchain based game integrating Groq API for dynamic storytelling and Solidity smart contracts for NFT collectibles
- Built full-stack system with React frontend, Flask backend, and Firebase for real-time player data and asset trading.

LLM From Scratch | *Python, PyTorch*

Mar 2025

- Built a transformer architecture from scratch including self-attention, tokenization, and training loop
- Tested generation capability on small corpora and validated syntactic structure

Fire & Smoke Detection System | *YOLOv8, OpenCV, Python*

Dec 2024

- Designed a real-time video stream detection system for smoke and fire hazards
- Utilized bounding box tracking and confidence score filters for false-positive reduction

ACHIEVEMENTS

Secured **Top 25 All India Rank** in HackHazard 2025 (Monad Track) for *QuestMint*, an AI + blockchain based game.

Achieved **Top 100 All India Rank** in HackHazard 2025 overall competition for the same project.

Secured **Top 10 position** at AMU Hacks 4.0 for *ChikitchakAI*, a healthcare AI assistant

TECHNICAL SKILLS

Languages: Python, C, C++, SQL (MySQL)

AI/ML Frameworks: PyTorch, TensorFlow, LangChain, Transformers, OpenCV

Tools & Platforms: Git, Docker, Jupyter, VS Code

Domains: Generative AI, LLMs, Computer Vision, Pretraining and Fine-Tuning models